FIITJEE

# JEE(Main)-2018

ALL INDIA TEST SERIES

## ANSWERS, HINTS & SOLUTIONS

## FULL TEST – II
## (Main)

| Q. No. | PHYSICS | Q. No. | CHEMISTRY | Q. No. | MATHEMATICS |
|---|---|---|---|---|---|
| 1. | **A** | 31. | **B** | 61. | **C** |
| 2. | **B** | 32. | **D** | 62. | **D** |
| 3. | **A** | 33. | **C** | 63. | **D** |
| 4. | **D** | 34. | **B** | 64. | **A** |
| 5. | **D** | 35. | **B** | 65. | **B** |
| 6. | **D** | 36. | **D** | 66. | **B** |
| 7. | **C** | 37. | **B** | 67. | **C** |
| 8. | **B** | 38. | **C** | 68. | **A** |
| 9. | **D** | 39. | **A** | 69. | **C** |
| 10. | **C** | 40. | **A** | 70. | **B** |
| 11. | **C** | 41. | **B** | 71. | **C** |
| 12. | **C** | 42. | **B** | 72. | **A** |
| 13. | **A** | 43. | **C** | 73. | **B** |
| 14. | **D** | 44. | **C** | 74. | **C** |
| 15. | **B** | 45. | **A** | 75. | **D** |
| 16. | **B** | 46. | **D** | 76. | **A** |
| 17. | **B** | 47. | **B** | 77. | **C** |
| 18. | **C** | 48. | **C** | 78. | **B** |
| 19. | **B** | 49. | **A** | 79. | **B** |
| 20. | **C** | 50. | **C** | 80. | **C** |
| 21. | **B** | 51. | **A** | 81. | **B** |
| 22. | **B** | 52. | **B** | 82. | **A** |
| 23. | **A** | 53. | **D** | 83. | **B** |
| 24. | **C** | 54. | **A** | 84. | **A** |
| 25. | **B** | 55. | **C** | 85. | **C** |
| 26. | **C** | 56. | **C** | 86. | **B** |
| 27. | **A** | 57. | **B** | 87. | **B** |
| 28. | **C** | 58. | **B** | 88. | **A** |
| 29. | **C** | 59. | **A** | 89. | **A** |
| 30. | **A** | 60. | **B** | 90. | **A** |

# Physics

## PART – I

### SECTION – A

1. A

$\tau = 0 \Rightarrow \alpha = 0 \Rightarrow \vec{\omega}$ = constant & $\vec{L} = I\vec{\omega}$ = constant

2. B

$\vec{v}$ = constant $\therefore\ \vec{a} = 0\ \therefore\ \sum\vec{F} = 10 - f = 0 \Rightarrow f = 10N$

3. A

The man must swim along the direction of shortest distance.

4. D

In the relative frame

$\vec{S}_0 = -h\hat{j},\quad \vec{u}_{rel} = v\cos\theta\,\hat{i} + (v\sin\theta - v)\hat{j};\quad \vec{a}_{rel} = 0$

5. D

for conservative force $F_x = \dfrac{-dU}{dx}; F_y = \dfrac{-dU}{dy}$

$\dfrac{\delta F_x}{\delta y} = \dfrac{\delta F_y}{\delta x} = \dfrac{\delta^2 U}{\delta x \delta y}$

6. D

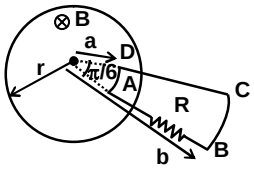
if $\mu_1 < \mu_2$, then also N > 0

7. C

8. B

$e = \dfrac{d\phi}{dt} = \dfrac{(r^2 - a^2)}{2}\dfrac{\pi}{6}\dfrac{dB}{dt}$

$\therefore i = \dfrac{(r^2 - a^2)\pi B_0}{12R}$




9. D

No current will flow through the capacitor, so charge will be zero.

10. C

$C = \dfrac{\in_0 A}{x}$, where x is separation between plates.

$\dfrac{1}{C}\dfrac{dC}{dT} = \dfrac{1}{A}\dfrac{dA}{dT} - \dfrac{1}{x}\dfrac{dx}{dT}$

For $\dfrac{dC}{dT} = 0$, $\dfrac{1}{x}\dfrac{dx}{dT} = \dfrac{1}{A}\dfrac{dA}{dT}$

$\Rightarrow\ \alpha_S = 2\alpha$

11. C

For pure translatory motion of object, the force should act at the centre of mass of the object

$y_{cm} = \dfrac{m \times 2\ell + 2m \times \ell}{3m} = \dfrac{4\ell}{3}$

12. C
Equipotential surfaces are coaxial cylinders, therefore (4, 3, 4) and (3, 4, 0) lie on equipotential surface.

13. A
Energy absorbed per unit time by earth's surface, if its radius is r, is

$$\dot{Q}_{abs} = \frac{\sigma 4\pi R_S^2 T_S^4}{4\pi R^2} \times \pi r^2$$

Energy radiated per unit time by earth's surface, $\dot{Q}_{rad} = \sigma(4\pi r^2)T^4$

In equilibrium,

$$\dot{Q}_{abs} = \dot{Q}_{rad}$$

$$\Rightarrow \quad T^4 = \frac{T_S^4}{4}\frac{R_S^2}{R^2}$$

14. D

15. B
CM will be at rest, m (50 – x) = 99 m (x)

$$\Rightarrow x = \frac{1}{2}m$$

16. B
$PV = m\Sigma v_x^2$

$$\Rightarrow m = \frac{100 \times 1}{10^{28}} = 10^{-26} \text{ kg} = 10^{-23} \text{ g}$$

17. B
Potential at P due to rings

$$V_P = \frac{\lambda 2\pi R}{4\pi \in_0 R} - \frac{\lambda 2\pi R}{4\pi \in_0 \sqrt{2}R} = \frac{\lambda}{2 \in_0}\left(1 - \frac{1}{\sqrt{2}}\right)$$

Potential at Q due to rings is

$$V_Q = -\frac{\lambda}{2 \in_0}\left(1 - \frac{1}{\sqrt{2}}\right)$$

$\Delta$(K.E.) + $\left(V_Q - V_P\right)q = 0$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{\lambda}{\in_0}\left(1 - \frac{1}{\sqrt{2}}\right)q$$

18. C
No e.m.f. will be induced in AC because electric lines of force are perpendicular to AC and e.m.f. induced in AB = e.m.f. induced in BC.

$$\therefore \text{ e.m.f induced in AB } = \frac{1}{2}\left|\frac{d\phi}{dt}\right| = \frac{1}{2}r^2\frac{dB}{dt}$$

19. B

$$T = \frac{mv^2}{R}, \text{ and } T = mg$$

20. C
Initially $\sigma\, 4\pi\, r_A^2 T_A^4 = \sigma 4\pi r_B^2 T_B^4$

Now $\frac{Q_A}{Q_B} = \frac{\sigma(4\pi)(r_A)^2(2T_A)^4}{\sigma 4\pi(2r_B)^2(T_B)^4} = \frac{4r_A^2T_A^4}{r_B^2T_B^4} = 4$

21. B
Net field at the centre of cube = 0
$\therefore \quad E_i + E_q = 0$
$\Rightarrow \quad E_i = -\dfrac{q}{4\pi \in_0 d^2}$
$\therefore \quad |E_i| = \dfrac{q}{4\pi \in_0 d^2}$

22. B

23. A
$-kbx = \dfrac{2Mb^2\alpha}{3}$
$\alpha = -\dfrac{3}{2}\dfrac{kx}{Mb}$

24. C
Use Snell's law: $\mu \sin \theta$ = constant

25. B
$mv^2/r = GMm/r^2$

26. C

27. A
$V = V_1 + V_2$ and $V\gamma = V_1\gamma_1 + V_2\gamma_2$

28. C
At first, the level of water will gradually rise to height $h_0$. After reaching height $h_0$, some of the water will get drained through the siphon. As soon as the entire cross section of the top of the siphon pipe is filled with water, the water level will begin to drop since, from given conditions, the flow-rate of the water flowing from pipe B is greater than from pipe A. As the water level drops, the pressure that drives the siphon decreases and as a result the velocity of water in pipe B decreases. Level will continue to sink until the flow rate of water through pipe B becomes equal to that through pipe A.

29. C
$T^2 \propto$ (semi-major axis)$^3$

30. A
Use $\dfrac{d\hat{e}_r}{dt} = \dfrac{d\theta}{dt}\hat{e}_\theta$ and $\dfrac{d\hat{e}_\theta}{dt} = -\dfrac{d\theta}{dt}\hat{e}_r$

# Chemistry

## PART – II

### SECTION – A

31. B

Subshell with lower $(n+\ell)$ value is lower in energy and energy is emitted in transition from a subshell of higher energy to lower energy subshell.

32. D

$r_1 = k[A]^n[B]^m$

$r_2 = k[2A]^n\left[\frac{B}{2}\right]^m = 2^{n-m}k[A]^n[B]^m$

$\frac{r_1}{r_2} = 2^{n-m}$

33. C

When a reaction is reversed, magnitude of enthalpy change remains same.

34. B




35. B

$p \alpha n$ at constant volume of temperature.

$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$

| | | |
|---|---|---|
| $2p$ | $p$ | $O$ |
| $2p-2x$ | $p-x$ | $2x$ |

$p_T = 2p-2x+p-x+2x = 1$

$3p-x=1 \ \& 2x = \frac{p-x}{4}$

Solving $x = \frac{1}{26} \& \ p = \frac{9}{26}$

$k_p = \frac{2(x)^2}{(p-x)(2p-2x)^2}$

36. D

$E \alpha \frac{1}{\lambda}$

Maximum wavelength radiation means minimum energy radiation which is emitted when transition occurs from $4 \longrightarrow 2$.

$\frac{1}{\lambda} = R\left(\frac{1}{4}-\frac{1}{16}\right)$

37. B

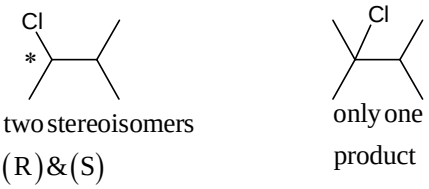
Given silicate anion is $\left[(SiO_3)^{2-}\right]_6$

$\therefore n = 12$

38. C
Chemisorption increases with increase in temperature initially and then decrases.

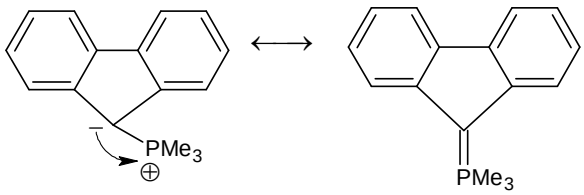
39. A
$$\left[I^-\right]=\sqrt{\frac{7.2\times10^{-9}}{0.005}}=1.2\times10^{-3}M$$
Maximum moles of NaI that can be added to 500 ml of $Pb\left(NO_3\right)_2$ are
$$\frac{1.2\times10^{-3}}{1000}\times500=6\times10^{-4}moles\Rightarrow0.09g$$

40. A
Equivalents of $Cr_2O_7^{2-}=0.136$
$$n_b\times n=0.136\Rightarrow n=\frac{0.136}{6}=0.226$$

41. B


Total products = 3

42. B
Oxidation state of all species remain unchanged so it is not redox

43. C


Negative charge can be delocalised on phosphorous also in addition to phenyl rings. So the anion is more stabilised by resonance.

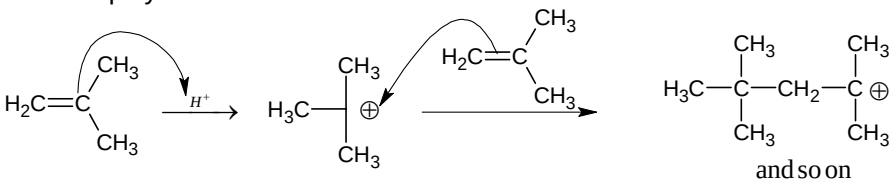
44. C
The substrate which can form most stable carbocation will give maximum racemisation

45. A
Epoxide undergo ring opening with bases. Greater the ring strain, greater is the reactivity towards base.

46. D
Protonated methanol is a stronger acid than methanol.

47. B
Reactivity of carbonyl compounds follow the order acid chlorides > aldehydes > ketones > ester

48. C
$OH^-$ Converts phenol to phenoxide ion which is strongly activating so attack occurs at ortho position w.r.t. hydroxyl group

49. A
Cationic polymerisation occurs as:


50. C

$\left[Cr\left(NH_3\right)_6\right]^{3+}$ - $\left(d^3\right)$-$d^2sp^3$ hybridised & paramagnetic

$\left[Co\left(CN\right)_6\right]^{3-}$ - $\left(d^6\right)$-$d^2sp^3$ hybridised ($CN^-$ is a strong field ligand)

51. A

Fact based

52. B

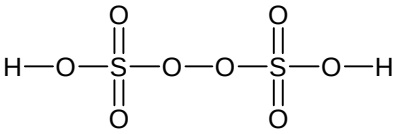
$ZnS$ is white coloured ppt which is soluble in acidic solution below $pH=6$

53. D

Aluminium is trivalent & $Be$ is divalent

54. A

$H_2SO_4$ on electrolysis forms $H_2S_2O_8$



55. C

Atomic radii increases for $Eu$ & $4b$ in lanthanoids series.

56. C

At constant volume $\frac{p_1}{T_1}=\frac{p_2}{T_2}\Rightarrow p_2=p_1\frac{T_2}{T_1}=\frac{3}{2}$

$\Rightarrow\ dG=Vdp-SdT$

$=V\Delta p-\int_{200k}^{300k} 2+10^{-T}T\,dT$

57. B

$\alpha=\sqrt{\frac{K_a}{C}}=\sqrt{\frac{1.6\times10^{-5}}{0.01}}$

$\lambda_m=\alpha\lambda_m^\circ=\frac{K\times1000}{\text{molarity}}$ when $\lambda_m$ is in $S\,cm^2mol^{-1}$

$G=\frac{K}{\text{cell constant}}$

58. B

$\rho=\frac{Z\times M}{N_A\times a^3}\Rightarrow a=\left(r_{Cd^{2+}}+r_{O^{2-}}\right)$

59. A

At Boyle's temperature, a real gas behaves approximately ideally for an appreciable range of pressure.

60. B

$Mn_xO_y+\frac{y}{2}C\longrightarrow\frac{y}{2}CO_2+x\,M_n$

$\frac{y}{2}=\frac{13.2}{44}\Rightarrow y=0.6$

$55x+0.6\times16=31.6\Rightarrow x=0.4\Rightarrow x:y=2:3$

# Mathematics

## PART – III

### SECTION – A

61. C

$$\sum_{n=1}^{5}\sin^{-1}\left(\sin\left(2n\pi-\frac{\pi}{6}\right)\right)+\sum_{n=1}^{5}\sin^{-1}\left(\sin(2n-1)\pi-\frac{\pi}{6}\right)$$

$$\sum_{n=1}^{5}\sin^{-1}\left(-\sin\frac{\pi}{6}\right)+\sum_{n=1}^{5}\sin^{-1}\left(\sin\frac{\pi}{6}\right)$$

$$=-\frac{5\pi}{6}+\frac{5\pi}{6}=0$$

62. D

$$\lim it=\frac{f'(1)}{2}=\frac{\pi-4}{16\sqrt{2}}$$

63. D

$\because x\geq\sin x\forall x\in[0,10\pi]$

$\Rightarrow x+2^{x}\geq\sin x+2^{\sin x}\forall x\in[0,10\pi]$

& equality holds only at x=0

64. A

65. B

Let $A(\vec{0}),B(\vec{a}),c(\vec{b})\&D(\vec{c})$

$$\Rightarrow G_1\left(\frac{\vec{a}+\vec{b}}{3}\right),G_2\left(\frac{\vec{b}+\vec{c}}{3}\right),G_3\left(\frac{\vec{c}+\vec{a}}{3}\right)$$

$$\Rightarrow \text{volume of } AG_1G_2G_3=\frac{1}{6}\left(\frac{\vec{a}+\vec{b}}{3}\frac{\vec{b}+\vec{c}}{3}\frac{\vec{c}+\vec{a}}{3}\right)$$

$$=\frac{2}{27}\frac{1}{6}\left[\vec{a}\vec{b}\vec{c}\right]=\frac{2}{27}\times 81$$

66. B

Let plane is

$(x+2y+3z-5)+\lambda(3x+2y+z-5)=0$

$\because$ plane is parallel to $\frac{x-1}{1}=\frac{y-2}{-1}=\frac{z-3}{1}$

$\Rightarrow 1(1+3\lambda)-1(2+2\lambda)+1(3+\lambda)=0$

$\Rightarrow \lambda=-1$

67. C

$A^{-1}+B^{-1}=2I$

$\Rightarrow I+AB^{-1}=2A$

$\Rightarrow B+A=2AB \text{ or } 2AB=I$

68. A

Required sum is coefficient of $x^{10}$ in the expansion of $(1+x)^{30}(1+x)^{20}(1+x)^{10}$ i.e ${}^{60}C_{10}$

69. C

70. B

$$\text{Area}=(2\alpha)\sin\left(\frac{\pi}{2}+\alpha\right)=2\alpha\cos\alpha$$

$$\frac{dA}{d\alpha}=2(-\alpha\sin\alpha+\cos\alpha)=2\sin\alpha(-\alpha+\cot\alpha$$

$$\frac{dA}{d\alpha}=0\Rightarrow\alpha=\cot\alpha\left\{\because\alpha\in\left(0,\frac{\pi}{2}\right)\right\}$$

71. C

$$\lim_{x\to0}\frac{f(x).\frac{x^2}{2}+x\int_0^x f(t)dt}{x^2}$$

$$=\frac{f(0)}{2}+\lim_{x\to0}\frac{\int_0^x f(t)dt}{x}=1+f(0)=3$$

72. A

$$f^{-1}(x)=g(x)\Rightarrow x=f(g(x))$$

$$\Rightarrow f'(g(x))g'(x)=1$$

$$\Rightarrow\int_1^3\left(g(x)+\frac{x}{f'(g(x))}\right)dx$$

$$=\int_1^3(g(x))dx=xg(x)\Big|_1^3=0$$

73. B

$$\int 2ydy=\int(1+x)dx\Rightarrow\frac{(x+1)^2}{2}-\frac{y^2}{1}=1$$

$$\Rightarrow e=\sqrt{1+\frac{1}{2}}=\sqrt{\frac{3}{2}}$$

74. C

Let A(asecθ,btanθ)

tangent at A is $\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$

$\because$ tangent passes through (0,-b) $\Rightarrow \tan\theta=1$

$$\therefore A\left(a\sqrt{2},b\right)$$

$$\Rightarrow CA\cos45^\circ=CM$$

$$\Rightarrow\sqrt{2a^2+4b^2}.\frac{1}{\sqrt{2}}=2b$$

$$\Rightarrow a^2=2b^2$$

75. D

$$(36,-24)\equiv(at_2^2,2at_2)\Rightarrow t_2=-3$$

$\because\ t_2 = -t - \frac{2}{t} \Rightarrow t^2 - 3t + 2 = 0 \Rightarrow t = 1 or\ 2$

Focal distance = $a + at^2 = 8 or\ 20$

76. A

Two parabolas opening upwards do not intersect, if they have same axis

$\Rightarrow -\frac{b}{2a} = \frac{-q}{2p}$ or aq=bp

$\Rightarrow$ maximum value is given by $(c-r)^2$

77. C

$n(A\times B)=9$

${}^9C_1 + {}^9C_3 + .... + {}^9C_9 = 2^8 = 256$

78. B

$P\left(\frac{\bar{A}}{B}\right) = \frac{2}{5} \Rightarrow \frac{P(\bar{A})P(B)}{P(B)} = \frac{2}{5}$

$\Rightarrow P(A) = \frac{3}{5}$

$P(A \cap B) = P(A)P(B) = \frac{3}{5}\left(\frac{3}{4} - \frac{3}{5}\right) = \frac{9}{100}$

79. B

$\frac{z + \bar{z}}{2} = |z - 1|$

$\Rightarrow x^2 = (x-1)^2 + y^2\ or\ y^2 = 2x - 1$

Let $z_1\left(\frac{t_1^2+1}{2}, t_1\right) \& z_2\left(\frac{t_2^2+1}{2}, t_2\right)$

$\because \arg(z_1 - z_2) = \frac{\pi}{3} \Rightarrow \frac{t_2 - t_1}{\frac{t_2^2 - t_1^2}{2}} = \sqrt{3}$

$\Rightarrow \mathrm{Im}(z_1 + z_2) = \frac{2}{\sqrt{3}}$

80. C

Locus of z is perpendicular bisector of line segment joining $\omega \& -\omega$

81. B

Let circle is $x^2 + y^2 - 2\alpha x - 2\beta y = 0$

$\because OA + 2OB = K \Rightarrow 2\alpha + 4\beta = k$

$\Rightarrow$ circle is $x^2 + y^2 - (k - 4\beta)x - 2\beta\gamma = 0$

$(x^2 + y^2 - kx) + 2\beta(2x - y) = 0$

$\Rightarrow$ Fixed point lies on $2x = y$

82. A

Area= $\frac{1}{4}$ area of circle $\frac{\pi}{2}$

83. B

$$\tan 30^\circ = \frac{AB}{3} \& \tan\theta = \frac{AB}{AQ}$$

$$\Rightarrow AB = \sqrt{3} \& \tan\theta = \frac{AB}{\sqrt{AP^2 + PQ^2}}$$

$$\Rightarrow \tan\theta = \frac{\sqrt{3}}{2\sqrt{3}} = \frac{1}{2}$$

84. A

Reflexive because $A^2 = A^2$

Symmetric because $AB = BA \Rightarrow BA = AB$

Not transitive because $AB = BA \ \& \ BC = CB \not\Rightarrow AC = CA$

85. C

86. B

$$f'(x) = 4x^3 + 3\lambda x^2 + 2x$$

$$\because f'\left(\frac{1}{2}\right) = 0 \Rightarrow \lambda = -2$$

$$\Rightarrow f'(x) = 2x(2x-1)(x-1)$$

Minimum value $= f(0) = f(1) = 0$

87. B

$$\bar{x} = \frac{\sum x_i}{100} = 0 \&$$

$$\frac{\sum |x_i - \bar{x}|}{100} = 5 \Rightarrow \sum_{1 \le i < j \le 100} |x_i x_j| = (500)^2$$

$$\Rightarrow \frac{\sum x_i^2}{100} = \frac{(500)^2 - 2\sum |x_i x_j|}{100} = 2500 - 1600$$

$$S.D. = \sqrt{\frac{\sum (x_i - \bar{x})^2}{100}} = \sqrt{900} = 30$$

88. A

$p \to q$ must be true & q→r must be false

∴ q must be true, r must be false but p can be true or false.

89. A

$\because f(x)y^2 + ay + g(x) = 0$ has real roots.

$$\therefore D \ge 0 \Rightarrow a^2 \ge 4 f(x) g(x)$$

$$\Rightarrow a^2 \ge 4 \sin x \cos x$$

$$\Rightarrow a^2 \ge 2 \sin 2x \cos x$$

$$\Rightarrow a \in \left(-\infty, -\sqrt{2}\right] \cup \left[\sqrt{2}, \infty\right)$$

90. A

$\because$ sum of two A.P's also an A.P.

$x_4+y_4+z_4+w_4=A+3D=8$ &

$x_{10}+y_{10}+z_{10}+w_{10}=A+9D=20$

$\Rightarrow A=2,D=2$

$\Rightarrow x_{20}+y_{20}+z_{20}+w_{20}=A+19D=40$

Now $\frac{x_{20}+y_{20}+z_{20}+w_{20}}{4} \geq \left(x_{20}.y_{20}.z_{20}.w_{20}\right)^{1/4}$

$\Rightarrow \left(x_{20}.y_{20}.z_{20}.w_{20}\right) \leq 10^4$